

Project Overview

Brainstorming creates an open space for creative problem solving. This session focuses on improving ServiceNow employee-training data imports, reference resolution, user roles, and ACL-based access control.

Project Name	Import Data using Transform Maps	Maximum Marks	4 Marks
Date	31 January 2025	Team ID	6ab615fddcdae33acb38af6f

Step 1: Team Gathering, Collaboration and Problem Statement

Bring the team together and define the central problem the session will address.

Project Description	A ServiceNow project that implements data import through Transform Maps, dot-walking, user roles, and ACL-based access control to securely manage employee training records.
Target Users	ServiceNow System Administrators and the HR Operations team, who own employee training imports, Transform Maps, reference fields, and field-level security.
Problem Statement	How might we import employee training data accurately into ServiceNow, resolve manager and department references through dot-walking, and protect sensitive training records with appropriate user roles and ACLs?
Desired Outcome	Reliable imports with no duplicate records, correct reference relationships, and controlled access to sensitive employee training information.

User Perspective: Empathy Map

Persona: System Administrator / HR Operations specialist responsible for training imports.

THINK & FEEL Worries about duplicate users, inaccurate mappings, and unauthorized access to employee training scores.	HEAR Leadership requires role-based access control. HR Operations reports missing IDs and inconsistent course names.
SEE Raw CSV files with non-standard IDs, and manual entry processes that introduce errors into training tracking.	SAY & DO Manually verifies staging records, configures field-level ACLs, and writes onBefore transform scripts for validation.
PAIN Duplicate training entries, broken reference linkages, and unauthorized visibility of confidential scores.	GAIN Accurate automated imports, dynamic manager and department resolution, and strong role-based data security.

Step 2: Brainstorm, Idea Listing and Grouping

Generate a broad set of possible solutions first, then group related ideas into technical themes.

Cluster	Idea	Description
Data Ingestion & Deduplication	Single Coalesce Key	Use employee ID as the Transform Map coalesce field to identify existing records.
	Compound Coalesce	Coalesce on employee ID plus course code, so one employee can hold multiple course records without unwanted duplicates.
	Duplicate Prevention Script	Use an onBefore transform script to stop an insert when a matching record already exists.
Reference & Dot-Walking	Native Reference Mapping	Map the employee reference to the sys_user table using an appropriate display value.
	Dynamic Dot-Walking	Resolve assigned_to.manager and assigned_to.department dynamically from the employee reference.
	Scripted Reference Lookup	Use a sys_user lookup script when standard reference matching is insufficient.
Roles & ACL Governance	Table-Level ACL	Restrict access to the employee training table to users holding the required application role.
	Field-Level Score ACL	Protect the sensitive score field with dedicated Read and Write ACL rules.
	Dynamic ACL	Use ACL logic to check whether the logged-in user is authorized through the employee/manager relationship.
Automation & Error Handling	Scheduled Import	Use a Scheduled Import Set for recurring employee training CSV imports.
	Error Notification	Trigger an event and notification when a Transform Map produces more errors than a defined threshold.

Total ideas generated: 11 across 4 themes. Each theme maps to one stage of the import-to-access lifecycle: ingest, resolve, secure, monitor.

Step 3: Idea Prioritization

Ideas are compared on importance/impact and feasibility to decide what moves into the implementation plan.

Priority Area	Idea	Reason for Placement	Implementation Focus
High Importance / High Feasibility	Compound Coalesce	Directly addresses duplicate records and is practical to configure in a Transform Map.	Coalesce on employee ID + course code.
High Importance / High Feasibility	Dynamic Dot-Walking	Directly addresses manager and department reference resolution.	Use assigned_to.manager and assigned_to.department.
High Importance / High Feasibility	Field-Level ACL	Directly protects sensitive employee training scores.	Apply Read/Write ACLs to the score field.
High Importance / Lower Feasibility	Dynamic ACL Script	Gives relationship-based access control but needs extra scripting and testing.	Validate the manager/employee relationship in ACL logic.
High Importance / Lower Feasibility	Automated Error Notification	Improves monitoring but requires event and notification configuration.	Notify after a defined transform error threshold.
Lower Importance / High Feasibility	Single Coalesce / Native Mapping	Simple approaches, but they cover less of the project requirements.	Use only where source data is already standardized.
Lower Importance / Lower Feasibility	Complex Purging / SFTP Integration	Adds complexity beyond the core import, reference, and access-control problem.	Keep outside the initial implementation scope.

Legend: green = implement first; amber = implement after core setup; grey = defer or use selectively.

Success Metrics

Measure	Target
Duplicate records after import	0 duplicates when the same CSV is imported twice
Reference accuracy	100% of rows resolve employee, manager, and department correctly
Access control	Users without the required role cannot read or write the score field
Error handling	Notification fires when errors exceed the configured threshold

Final Implementation Roadmap

Sequenced from highest-priority ideas, with a suggested owner and exit check for each stage.

#	Stage	Suggested Owner	Exit Check
1	Configure Transform Map coalesce	System Administrator	Re-running the same file creates no new records
2	Configure reference mapping and dot-walking	System Administrator	Manager and department populate for all test rows
3	Create user roles and table/field ACLs	Security / ACL Administrator	Unauthorized test user is denied on table and score field
4	Test staging and transformed records	HR Operations + Tester	Staging and target counts reconcile; no unexplained errors
5	Validate access and error handling	Tester + HR Operations	Role tests pass; error notification received at threshold

Risks and Mitigations

Risk	Mitigation
Non-standard or missing employee IDs in the CSV	Clean and validate IDs before import; reject blank keys in an onBefore script.
Inconsistent course names or codes	Standardize against a reference list; log unmatched values in staging for review.
Manager or department blank on the user record	Check sys_user data completeness first; flag rows with empty dot-walked values.
ACL misconfiguration exposes or blocks scores	Test with at least one authorized and one unauthorized user before go-live.
Dynamic ACL script adds complexity	Deliver table and field ACLs first; add the relationship check as a second iteration.

Summary

The team will prioritize a compound coalesce key, dynamic dot-walking, and field-level ACLs as the core of the solution. Dynamic ACL logic and error notifications follow once the core import is stable, while purging and SFTP integration stay out of the initial scope.